Report

Of

SmartBranch 360

Submitted

To

School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of

B.Tech CSE

By

Roll Number of Student: 2410031226

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2026-27

CANDIDATE'S DECLARATION

I, Richa Goswami do hereby declare that the internship report titled **"SmartBranch 360"** has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. All the references have been quoted and I have not taken as such any material from any other source. I affirm to you that this report is my own and has not been submitted to any other institute for any degree/diploma requirement and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature
Student Name Richa Goswami
Roll No. 2410031226

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluably constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I express my gratitude to my parents for their blessings.

Signature

Student Name Richa Goswami

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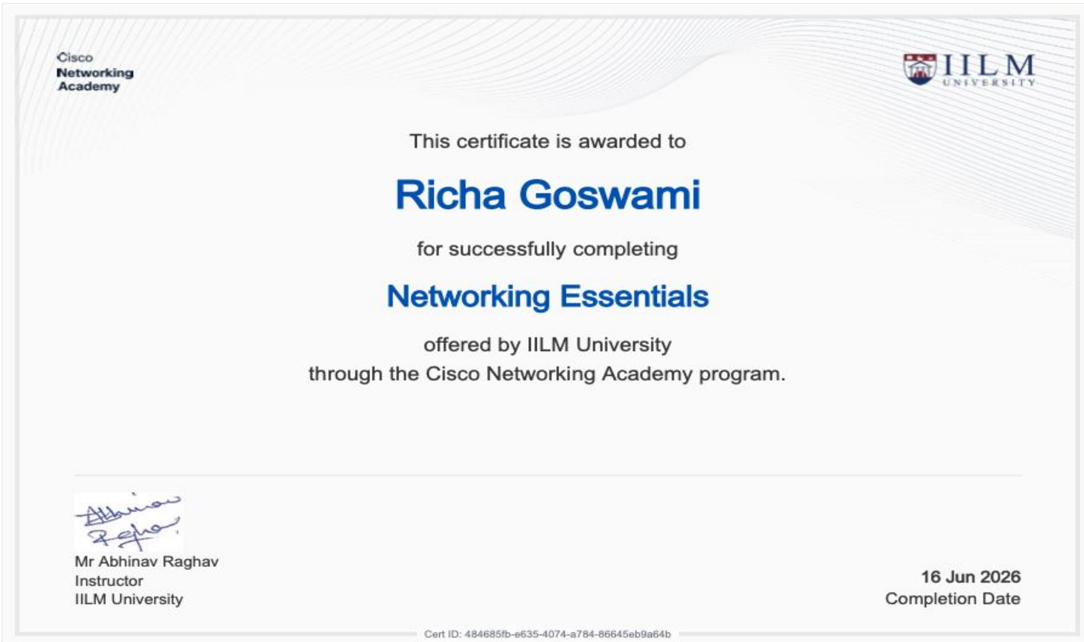
INTERNSHIP COMPLETION CERTIFICATE

The following certificates, issued by IILM University through the Cisco Networking Academy program, are attached in evidence of the internship/training modules completed as part of this project work.



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Instructor
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Completion Date

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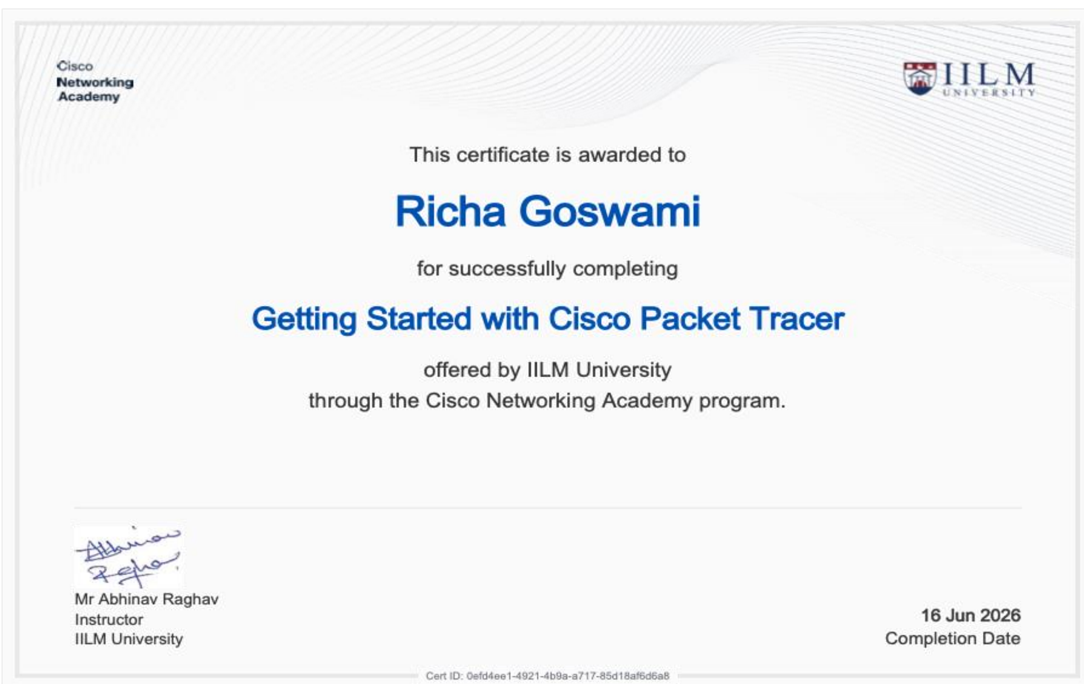

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Project Description

4.1 Introduction

The network project aims to create a safe and effective enterprise network with the help of Cisco Packet Tracer. The proposed network is subdivided into several Virtual Local Area Networks (VLANs) in order to serve different categories of users and devices simultaneously. The network consists of Employee, Guest, Server, and Management VLANs.

For inter-VLAN communication router-on-a-stick technique is used so that the router could serve as a gateway for every VLAN. The Dynamic Host Configuration Protocol (DHCP) is utilized for the purpose of IP addresses receiving by the devices in the network automatically. Access Control Lists (ACLs) are deployed in order to block Guest users from accessing the Employee, Server, and Management networks.

The project illustrates the basic principles of networking technology, such as VLAN segmentation, trunking, inter-VLAN routing, DHCP, network security, and testing connectivity. Cisco Packet Tracer is employed for the design, configuration and testing of the network topology created.

4.2 Organization Profile

The proposed network architecture is for a medium-sized company featuring diverse departments and network user types. This organization needs a trustworthy, organized and secure network setup to accommodate employees, servers, visitors and network management tasks.

The network plan organizes the units into separate VLANs according to the company needs. Employee computers belong to the Employee VLAN. Guest equipment is connected to the Guest VLAN, key services are hosted in the Server VLAN, and network administration computers are part of the Management VLAN.

The design supports controlled communications between various departments and safeguards classified company information against unauthorized access.

4.3 Problem Statement

A flat network, with all devices on the same broadcast domain, can result in issues with security, performance, and management. For instance, external users could gain access to the internal resources of the organization, while there may be excessive broadcast traffic.

Thus, the organization needs to have some network infrastructure that guarantees logical separation of users and services, controlled communication between different segments of the network, automatic allocation of IP addresses, and improvement in security.

In this project, the requirements are met by having VLAN segmentation, inter-VLAN routing, DHCP services, trunk links, and Access Control Lists to segregate the Guest network from other internal networks.

4.4 Project Objectives

- To design and implement a structured enterprise network using Cisco Packet Tracer.
- To create separate VLANs for Employees, Guests, Servers, and Management devices.
- To configure trunk links for carrying multiple VLANs between network devices.
- To implement inter-VLAN routing using router-on-a-stick.
- To configure DHCP for automatic IP address assignment.
- To implement Access Control Lists (ACLs) for network security.
- To prevent Guest users from accessing Employee, Server, and Management networks.
- To allow authorized communication between required network segments.
- To provide centralized network management through a dedicated Management VLAN.
- To test and verify connectivity and security across the entire network.

4.5 Scope of the Project

The scope of this project includes the design, configuration, and testing of a small enterprise network in Cisco Packet Tracer.

The project covers:

- Creation and configuration of VLAN 10 for Employees.
- Creation and configuration of VLAN 20 for Guests.
- Creation and configuration of VLAN 30 for Servers.
- Creation and configuration of VLAN 99 for Management.
- Configuration of access ports for end devices.
- Configuration of trunk ports between switches and the router.
- Configuration of router subinterfaces for inter-VLAN routing.
- Configuration of DHCP services for automatic IP addressing.
- Configuration of DHCP relay using the `ip helper-address` command where required.
- Implementation of ACL-based Guest network isolation.
- Verification of connectivity using ping and network diagnostic commands.
- Testing of communication between authorized and unauthorized network segments.

The project is implemented in a simulated environment and does not represent a physical deployment of networking equipment.

4.6 Technologies and Tools Used

This project makes use of Cisco Packet Tracer to implement and design topology. This program allows for simulation and testing as well as configuration of devices. Cisco routers were chosen for inter-VLAN routing, default gateway services, DHCP relay, and access control.

On the other hand, Cisco switches were utilized to create VLANs, configure access ports, and perform trunking between networking equipment.

Moreover, VLANs were applied to separate the network into Employee, Guest, Server, and Management sections logically. The IEEE 802.1Q standard technology was utilized to transmit numerous VLANs over trunk links. Router-on-a-Stick method was adapted through router subinterfaces to connect different VLANs to one another.

As far as DHCP services are concerned, this technology was utilized to provide and configure client devices with IP addresses, subnet masks, and default gateways automatically. DHCP relay allowed forwarding DHCP requests between subnets and the centralized DHCP server.

Lastly, in terms of security-related issues, Extended Access Control Lists (ACLs) were applied to control the movement of data packets and segment the Guest VLAN from more important Employee, Server, and Management networks.

The network connectivity was tested by utilizing the Ping utility to check the connectivity status for devices throughout the network. The Cisco Command-Line Interface (CLI) was employed to assess the condition of the switches and routers in the network, through the execution of commands including show vlan brief, show interfaces trunk, show ip interface brief, show access-lists and show ip route.

4.7 System Architecture

The architecture of the network is hierarchical and segmented. There is a Cisco router for the purpose of networking different VLANs, while switches connect end devices to the network.

The main segments of the network are:

- VLAN 10 – EMPLOYEE: The network used by employees for networking their computers.
  - Network: 192.168.10.0/24
  - Gateway: 192.168.10.1
- VLAN 20 – GUEST: The network used by guest computers.
  - Network: 192.168.20.0/24
  - Gateway: 192.168.20.1
- VLAN 30 – SERVER: The network used for servers.
  - Network: 192.168.30.0/24
  - Gateway: 192.168.30.1
- VLAN 99 – MANAGEMENT: The network used for network management.
  - Network: 192.168.99.0/24
  - Gateway: 192.168.99.1

The router is connected to the switch through a trunk link using IEEE 802.1Q encapsulation. The router uses sub-interfaces to handle routing for VLANs 10, 20, 30, and 99.

The guest VLAN uses a special ACL named GUEST_ISOLATION, which restricts guest VLAN traffic from accessing the employee, server, and management networks.

The overall architecture can be represented as:

End Devices → Access Switches → Trunk Link → Router → Inter-VLAN Routing → Controlled Network Resources

4.8 Methodology

The project was implemented using a systematic network design and configuration methodology.

Step 1: Network Planning

The required departments, devices, VLANs, IP networks, gateways, and security requirements were identified before configuration.

Step 2: VLAN Creation

Separate VLANs were created for Employees, Guests, Servers, and Management. Each VLAN was assigned a unique VLAN ID and IP subnet.

Step 3: Switch Configuration

Switch access ports were assigned to the appropriate VLANs. Trunk ports were configured to carry multiple VLANs between switches and the router.

Step 4: Inter-VLAN Routing

Router sub interfaces were configured on the router's GigabitEthernet interface. Each sub interface was assigned an IP address corresponding to its VLAN and configured with 802.1Q encapsulation.

Step 5: DHCP Configuration

DHCP services were configured to provide automatic IP addressing to client devices. DHCP relay was configured where necessary so that clients in different VLANs could communicate with the centralized DHCP server.

Step 6: Security Configuration

An extended Access Control List named GUEST_ISOLATION was configured. It prevents Guest VLAN traffic from accessing the Employee, Server, and Management networks.

Step 7: Connectivity Testing

The configuration was verified using commands such as show vlan brief, show interfaces trunk, show ip interface brief, show access-lists, and show ip route. End-to-end connectivity was tested using the `ping` command.

Step 8: Final Verification

The network was checked to ensure that authorized communication worked correctly and that restricted Guest-to-internal-network communication was blocked.

4.9 Expected Outcomes

The expected results of the project are:

1. A well-organized enterprise network.

2. Successful segregation of employee, visitor, server, and administrative traffic using VLANs.

3. Successful communication between the VLANs where it is allowed.

4. Automatic assignment of IP addresses to clients by using the DHCP service.

5. Successful trunking among the networking hardware.

6. Controlled access to the internal network through the ACL system.

7. Prevention of visitors accessing the employee's network, server, and management.

8. Improved safety of the network through logical separation and filtering of the traffic.

9. Simplified management of the network with the dedicated management VLAN.

10. Successful validation of the complete setup in Cisco Packet Tracer.

4.10 Certificates of Completion

The certificates for the modules I completed as part of this course track – Networking Essentials, Getting Started with Cisco Packet Tracer, and Exploring Networking with Cisco Packet Tracer are already attached earlier in this report, under Chapter 3 (Internship Completion Certificate).

5. Bibliography/References

1. Cisco Networking Academy. Introduction to Networks (ITN). Cisco Systems, Inc.
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5. RFC 2131. Dynamic Host Configuration Protocol (DHCP). Internet Engineering Task Force (IETF).
6. IEEE. IEEE 802.1Q – Virtual Bridged Local Area Networks. Institute of Electrical and Electronics Engineers.